Assignment 5 Solutions
Modern Cryptography (2)

Problem 1. Let $x, x' \in \{0,1\}^m$ be a collision for H_A, i.e.,

$$x \neq x' \qquad \text{and} \qquad H_A(x) = H_A(x').$$

By definition of H_A,

$$Ax \equiv Ax' \pmod{q}.$$

Hence

$$A(x - x') \equiv 0 \pmod{q}.$$

Define

$$z := x - x'.$$

Then $z \neq 0$ since $x \neq x'$, and every coordinate of z belongs to $\{-1, 0, 1\}$. Therefore

$$Az \equiv 0 \pmod{q},$$

so z is a nonzero short solution to the homogeneous system modulo q.

Moreover,

$$\|z\|_\infty \leq 1,$$

and hence also

$$\|z\|_2 \leq \sqrt{m}.$$

Thus z is a valid solution to the SIS problem (for example, with bound $\beta = \sqrt{m}$ in Euclidean norm, or $\beta = 1$ in ℓ_∞-norm).

Therefore, any algorithm that finds a collision for H_A yields an algorithm that solves SIS.

Problem 2. We prove EUF-CMA security in the random oracle model by reduction to ISIS.

Assume there exists a probabilistic polynomial-time adversary $\mathcal{A}$ that wins the EUF-CMA game against the scheme with non-negligible probability ε. We construct a probabilistic polynomial-time algorithm $\mathcal{B}$ that solves $\text{ISIS}_{n,m,q,\beta}$ with non-negligible probability.

Input to $\mathcal{B}$. $\mathcal{B}$ receives an ISIS instance $(A, u^\star)$, where

$$A \in \mathbb{Z}_q^{n \times m}, \qquad u^\star \in \mathbb{Z}_q^n.$$

Its goal is to find $\sigma^\star \in \mathbb{Z}^m$ such that

$$A\sigma^\star \equiv u^\star \pmod{q} \qquad \text{and} \qquad \|\sigma^\star\| \leq \beta.$$

Setup. $\mathcal{B}$ gives the public key A to $\mathcal{A}$.

Let q_H be an upper bound on the number of random-oracle queries made by $\mathcal{A}$. Algorithm $\mathcal{B}$ guesses an index

$$j \in \{1, \ldots, q_H\}$$

uniformly at random. Intuitively, $\mathcal{B}$ hopes that the j-th random-oracle query of $\mathcal{A}$ will be the message that is eventually forged.

$\mathcal{B}$ maintains a table for messages. For each message μ, the table stores

$$(\mu,\ H(\mu),\ \sigma_\mu).$$

Random oracle simulation. Whenever $\mathcal{A}$ queries $H(\mu)$:

- If μ is already in the table, return the stored value.
- Otherwise, if this is the j-th random-oracle query, define

  $$H(\mu) := u^\star,$$

  store $(\mu, u^\star, \perp)$, and return $u^\star$.
- Otherwise, sample a short vector $\sigma_\mu \in \mathbb{Z}^m$ such that $\|\sigma_\mu\| \le \beta$, define

  $$H(\mu) := A\sigma_\mu \bmod q,$$

  store $(\mu, H(\mu), \sigma_\mu)$, and return $H(\mu)$.

Signing oracle simulation. When $\mathcal{A}$ asks for a signature on a message μ:

- If μ has not yet been queried to the random oracle, then $\mathcal{B}$ first queries the random oracle on behalf of $\mathcal{A}$ and creates the corresponding table entry as above.
- If μ is the specially programmed message whose hash is $u^\star$, then $\mathcal{B}$ aborts.
- Otherwise, the table contains a short vector σ_μ such that

  $$A\sigma_\mu \equiv H(\mu) \pmod q \qquad \text{and} \qquad \|\sigma_\mu\| \le \beta.$$

  Then $\mathcal{B}$ returns σ_μ as the signature.

This simulation is perfectly consistent: every returned signature verifies correctly.

Forgery extraction. Eventually $\mathcal{A}$ outputs a forgery $(\mu^\star, \sigma^\star)$ such that:

- $\mu^\star$ was never queried to the signing oracle, and
- verification accepts, i.e.,

  $$A\sigma^\star \equiv H(\mu^\star) \pmod q \qquad \text{and} \qquad \|\sigma^\star\| \le \beta.$$

We may assume, except with negligible probability, that $\mu^\star$ was queried to the random oracle. Indeed, if $\mu^\star$ was never queried to H, then $H(\mu^\star)$ is a fresh uniform random value in $\mathbb{Z}_q^n$, and the probability that $\mathcal{A}$ can output a valid $\sigma^\star$ satisfying

$$A\sigma^\star \equiv H(\mu^\star) \pmod q$$

is negligible.

Therefore, except with negligible probability, $\mu^\star$ appears among the q_H random-oracle queries. If $\mathcal{B}$ guessed correctly, namely if $\mu^\star$ is the j-th random-oracle query, then

$$H(\mu^\star) = u^\star.$$

Since the forgery verifies, we have

$$A\sigma^\star \equiv H(\mu^\star) = u^\star \pmod q \qquad \text{and} \qquad \|\sigma^\star\| \le \beta.$$

Hence $\sigma^\star$ is a valid solution to the given ISIS instance, and $\mathcal{B}$ outputs $\sigma^\star$.

Success probability. If $\mathcal{A}$ forges with probability ε, then, except for a negligible loss, the forgery message appears among the q_H random-oracle queries. Since $\mathcal{B}$ guesses the correct query index with probability $1/q_H$, it follows that

$$\Pr[\mathcal{B} \text{ solves ISIS}] \ge \frac{\varepsilon - \mathsf{negl}(\lambda)}{q_H}.$$

Therefore, if $\mathcal{A}$ has non-negligible forging probability, then $\mathcal{B}$ solves $\mathrm{ISIS}_{n,m,q,\beta}$ with non-negligible probability, contradicting the assumed hardness of ISIS.

Hence the signature scheme is EUF-CMA secure in the random oracle model.

Problem 3.

1(a) We show that $f(x) = x^2 + 1$ is irreducible over $\mathbb{F}_3$.

A quadratic polynomial over a field is reducible if and only if it has a root in that field. We evaluate:

$$f(0) = 0^2 + 1 = 1 \neq 0,$$
$$f(1) = 1^2 + 1 = 2 \neq 0 \pmod 3,$$
$$f(2) = 2^2 + 1 = 4 + 1 = 5 \equiv 2 \neq 0 \pmod 3.$$

So $f(x)$ has no root in $\mathbb{F}_3$, and therefore it is irreducible over $\mathbb{F}_3$.

1(b) Since $x^2 + 1$ is irreducible, the quotient

$$\mathbb{F}_9 \cong \mathbb{F}_3[x]/(x^2+1)$$

is a field with $3^2 = 9$ elements.

Let

$$\alpha := x + (x^2+1).$$

Then in $\mathbb{F}_9$ we have

$$\alpha^2 + 1 = 0, \qquad \text{so} \qquad \alpha^2 = -1 = 2 \in \mathbb{F}_3.$$

Every residue class modulo $x^2 + 1$ has a unique representative of degree < 2, so every element of $\mathbb{F}_9$ can be written uniquely as

$$a + b\alpha, \qquad a, b \in \mathbb{F}_3.$$

Thus the nine elements are

$$0,\ 1,\ 2,\ \alpha,\ 1+\alpha,\ 2+\alpha,\ 2\alpha,\ 1+2\alpha,\ 2+2\alpha.$$

1(c) We compute the inverse of every nonzero element.

First,

$$1^{-1} = 1, \qquad 2^{-1} = 2.$$

Since $\alpha^2 = 2$, we have

$$\alpha \cdot (2\alpha) = 2\alpha^2 = 2 \cdot 2 = 4 \equiv 1,$$

so

$$\alpha^{-1} = 2\alpha, \qquad (2\alpha)^{-1} = \alpha.$$

For general $a + b\alpha$ with $a, b \in \mathbb{F}_3$, note that

$$(a + b\alpha)(a - b\alpha) = a^2 - b^2\alpha^2 = a^2 - 2b^2.$$

Hence, whenever $a + b\alpha \neq 0$,

$$(a + b\alpha)^{-1} = \frac{a - b\alpha}{a^2 - 2b^2}.$$

Using this formula:

$$(1 + \alpha)^{-1} = \frac{1 - \alpha}{1 - 2} = \frac{1 + 2\alpha}{2} = 2 + \alpha,$$

so

$$(1 + \alpha)^{-1} = 2 + \alpha, \qquad (2 + \alpha)^{-1} = 1 + \alpha.$$

Similarly,

$$(1 + 2\alpha)^{-1} = \frac{1 - 2\alpha}{1 - 2 \cdot 4} = \frac{1 + \alpha}{1 - 8} = \frac{1 + \alpha}{0}$$

It is easier to compute directly:

$$(1 + 2\alpha)(2 + 2\alpha) = 2 + 2\alpha + 4\alpha + 4\alpha^2 = 2 + 6\alpha + 4 \cdot 2 = 2 + 0 + 8 = 10 \equiv 1.$$

Hence

$$(1 + 2\alpha)^{-1} = 2 + 2\alpha, \qquad (2 + 2\alpha)^{-1} = 1 + 2\alpha.$$

Therefore the inverses are:

$$\begin{aligned} 1^{-1} &= 1, & 2^{-1} &= 2, \\ \alpha^{-1} &= 2\alpha, & (2\alpha)^{-1} &= \alpha, \\ (1 + \alpha)^{-1} &= 2 + \alpha, & (2 + \alpha)^{-1} &= 1 + \alpha, \\ (1 + 2\alpha)^{-1} &= 2 + 2\alpha, & (2 + 2\alpha)^{-1} &= 1 + 2\alpha. \end{aligned}$$

1(d) The multiplicative group $\mathbb{F}_9^\times$ has order 8, so its generators are precisely the elements of order 8.

We compute powers of α:

$$\alpha^2 = 2, \qquad \alpha^4 = 2^2 = 4 \equiv 1.$$

So α has order 4, not 8.

Now consider $1 + \alpha$:

$$(1 + \alpha)^2 = 1 + 2\alpha + \alpha^2 = 1 + 2\alpha + 2 = 2\alpha,$$

$$(1 + \alpha)^4 = (2\alpha)^2 = 4\alpha^2 = \alpha^2 = 2,$$

$$(1 + \alpha)^8 = 2^2 = 1.$$

Also,

$$(1 + \alpha)^4 = 2 \neq 1,$$

so the order of $1+\alpha$ is 8. Hence $1+\alpha$ is a generator.

Since $\mathbb{F}_9^\times$ is cyclic of order 8, the generators are the powers $(1+\alpha)^k$ with $\gcd(k,8)=1$, namely $k=1,3,5,7$. These are

$$1+\alpha, \quad (1+\alpha)^3=(1+\alpha)(2\alpha)=2+2\alpha, \quad (1+\alpha)^5=(1+\alpha)\cdot 2=2+2\alpha?$$

It is cleaner to list the order-8 elements directly:

$$1+\alpha, \qquad 2+\alpha, \qquad 1+2\alpha, \qquad 2+2\alpha.$$

These are exactly the generators of $\mathbb{F}_9^\times$.

2(a) Let

$$\mathbb{F}_8 \cong \mathbb{F}_2[x]/(x^3+x+1), \qquad \alpha := x+(x^3+x+1).$$

Then

$$\alpha^3+\alpha+1=0, \qquad \text{so} \qquad \alpha^3=\alpha+1.$$

Now compute:

$$\begin{gathered}\alpha^4=\alpha\cdot\alpha^3=\alpha(\alpha+1)=\alpha^2+\alpha,\\ \alpha^5=\alpha\cdot\alpha^4=\alpha(\alpha^2+\alpha)=\alpha^3+\alpha^2=(\alpha+1)+\alpha^2=\alpha^2+\alpha+1,\\ \alpha^6=\alpha\cdot\alpha^5=\alpha(\alpha^2+\alpha+1)=\alpha^3+\alpha^2+\alpha=(\alpha+1)+\alpha^2+\alpha=\alpha^2+1,\\ \alpha^7=\alpha\cdot\alpha^6=\alpha(\alpha^2+1)=\alpha^3+\alpha=(\alpha+1)+\alpha=1.\end{gathered}$$

2(b) Since

$$\alpha^7=1$$

and none of

$$\alpha,\alpha^2,\alpha^3,\alpha^4,\alpha^5,\alpha^6$$

equals 1, the multiplicative order of α is

$$\operatorname{ord}(\alpha)=7.$$

3 Let

$$G=\mathbb{F}_q^\times.$$

Then G is a finite abelian group of order $q-1$.

Let

$$m=\operatorname{lcm}\{\operatorname{ord}(a): a\in G\},$$

the exponent of G. Then every $a\in G$ satisfies

$$a^m=1.$$

So every element of G is a root of the polynomial

$$X^m-1.$$

Since G has $q-1$ elements, X^m-1 has at least $q-1$ roots in $\mathbb{F}_q$. A nonzero polynomial of degree m over a field has at most m roots, so

$$q-1\le m.$$

On the other hand, by Lagrange's theorem, for every $a\in G$,

$$\operatorname{ord}(a) \mid |G|=q-1.$$

Hence their least common multiple also divides $q-1$:

$$m \mid (q-1).$$

Therefore

$$m \le q-1.$$

Combining the two inequalities gives

$$m = q-1.$$

Now write

$$m = \prod_{i=1}^{r} p_i^{e_i}$$

as the prime factorization of m. Since m is the least common multiple of the orders of all elements of G, for each i there exists some $a_i \in G$ such that $p_i^{e_i}$ divides $\operatorname{ord}(a_i)$. Define

$$b_i := a_i^{\operatorname{ord}(a_i)/p_i^{e_i}}.$$

Then b_i has order exactly $p_i^{e_i}$.

Since G is abelian and the numbers $p_i^{e_i}$ are pairwise coprime, the element

$$b := b_1 b_2 \cdots b_r$$

has order

$$\operatorname{ord}(b) = \prod_{i=1}^{r} p_i^{e_i} = m = q-1.$$

Thus G contains an element of order $q-1$, so G is cyclic.

Therefore $\mathbb{F}_q^\times$ is cyclic.

4 We prove that $\mathbb{F}_{p^n}$ contains a subfield of size p^d if and only if $d \mid n$.

($\Rightarrow$) Suppose $K \subseteq \mathbb{F}_{p^n}$ is a subfield with $|K| = p^d$. Then $\mathbb{F}_{p^n}$ is a finite extension of K, so

$$[\mathbb{F}_{p^n} : K] = r$$

for some positive integer r. Since

$$[\mathbb{F}_{p^n} : \mathbb{F}_p] = [\mathbb{F}_{p^n} : K]\,[K : \mathbb{F}_p],$$

we get

$$n = rd.$$

Hence $d \mid n$.

($\Leftarrow$) Suppose $d \mid n$. Consider the set

$$K := \{x \in \mathbb{F}_{p^n} : x^{p^d} = x\}.$$

This is exactly the set of roots of the polynomial

$$X^{p^d} - X.$$

Since every root set of $X^{p^d} - X$ in an extension field forms a field, K is a subfield of $\mathbb{F}_{p^n}$.

Moreover, the polynomial $X^{p^d} - X$ has at most p^d roots, and over an algebraic closure its roots are precisely the elements of $\mathbb{F}_{p^d}$. Since $d \mid n$, the field $\mathbb{F}_{p^d}$ embeds into $\mathbb{F}_{p^n}$, so K has exactly p^d elements.

Therefore $\mathbb{F}_{p^n}$ contains a subfield of size p^d.

5 The subfields of $\mathbb{F}_{2^{16}}$ correspond exactly to the divisors of 16. The positive divisors of 16 are

$$1, 2, 4, 8, 16.$$

Hence the possible subfield sizes are

$$2^1,\ 2^2,\ 2^4,\ 2^8,\ 2^{16},$$

that is,

$$\boxed{2,\ 4,\ 16,\ 256,\ 65536.}$$

Problem 4.

1(a) The generator matrix is

$$G = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \end{pmatrix}.$$

Since the message space is $\mathbb{F}_2^2$, there are four codewords:

$$(0,0)G = (0,0,0,0,0),$$

$$(1,0)G = (1,0,1,0,1),$$
$$(0,1)G = (0,1,1,1,0),$$
$$(1,1)G = (1,1,0,1,1).$$

So the codewords are

$$\boxed{00000,\ 10101,\ 01110,\ 11011.}$$

1(b) The code length is $n = 5$, and the dimension is $k = 2$.
The nonzero codewords have weights

$$\mathrm{wt}(10101) = 3, \qquad \mathrm{wt}(01110) = 3, \qquad \mathrm{wt}(11011) = 4.$$

Hence the minimum distance is

$$d = 3.$$

Therefore the parameters are

$$\boxed{[5, 2, 3].}$$

1(c) Write

$$G = (I_2 \mid P), \qquad P = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Then a parity-check matrix is

$$H = (P^T \mid I_3) = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

1(d) For

$$r = (1, 1, 0, 1, 1),$$

we compute

$$Hr^T = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Thus the syndrome is

$$\boxed{Hr^T = (0, 0, 0)^T.}$$

2(a) Now

$$G = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 1 & 1 \end{pmatrix}$$

over $\mathbb{F}_3$. Since the rank is 2, the code has

$$|C| = 3^2 = 9$$

codewords.

2(b) The message space is $\mathbb{F}_3^2$. Let $(a, b) \in \mathbb{F}_3^2$. Then

$$(a, b)G = (a, b, a + b, 2a + b).$$

Enumerating all pairs (a, b) gives:

$$(0, 0) \mapsto (0, 0, 0, 0),$$

$$(1, 0) \mapsto (1, 0, 1, 2),$$
$$(2, 0) \mapsto (2, 0, 2, 1),$$
$$(0, 1) \mapsto (0, 1, 1, 1),$$
$$(0, 2) \mapsto (0, 2, 2, 2),$$
$$(1, 1) \mapsto (1, 1, 2, 0),$$
$$(1, 2) \mapsto (1, 2, 0, 1),$$
$$(2, 1) \mapsto (2, 1, 0, 2),$$
$$(2, 2) \mapsto (2, 2, 1, 0).$$

2(c) The minimum distance of a linear code equals the minimum Hamming weight among nonzero codewords.

All nonzero codewords listed above have weight at least 3, and for example

$$(1, 1, 2, 0)$$

has weight 3. Hence

$$d = 3.$$

2(d) A code with minimum distance d can correct up to

$$t = \left\lfloor \frac{d-1}{2} \right\rfloor$$

errors. Here

$$t = \left\lfloor \frac{3-1}{2} \right\rfloor = 1.$$

Therefore this code can correct one error.

3 Let C be an $[n,k,d]$ linear code over $\mathbb{F}_q$.

Delete any $d-1$ coordinates from every codeword. I claim that the resulting shortened vectors are all distinct. Indeed, if two different codewords became equal after deleting $d-1$ coordinates, then they differed in at most $d-1$ positions, contradicting the fact that the minimum distance is d.

Thus after deleting $d-1$ coordinates, we still have q^k distinct vectors of length

$$n-(d-1) = n-d+1.$$

But the total number of vectors of that length over $\mathbb{F}_q$ is

$$q^{n-d+1}.$$

Therefore

$$q^k \le q^{n-d+1},$$

which implies

$$k \le n-d+1.$$

Rearranging gives the Singleton bound:

$$\boxed{d \le n-k+1.}$$

4(a) The matrix is

$$G = \begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

Observe that over $\mathbb{F}_2$,

$$(1,1,0,1,0) + (1,0,1,1,1) = (0,1,1,0,1),$$

so the third row equals the sum of the first two. Hence the rank is 2, and one row is redundant.

Perform row operations:

$$R_2 \leftarrow R_2 + R_1,$$

giving

$$\begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

Then

$$R_3 \leftarrow R_3 + R_2,$$

giving

$$\begin{pmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Finally,

$$R_1 \leftarrow R_1 + R_2,$$

giving

$$G' = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}.$$

Thus a systematic generator matrix is

$$\boxed{G' = (I_2 \mid P) = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \end{pmatrix}.}$$

4(b) Here

$$P = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}, \qquad P^T = \begin{pmatrix} 1 & 1 \\ 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

So a corresponding parity-check matrix is

$$H = (P^T \mid I_3) = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{pmatrix}.$$

4(c) We verify that

$$G'H^T = 0.$$

Indeed,

$$H^T = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Multiply the first row of G' by H^T:

$$(1,0,1,1,1)H^T = (1,1,1) + (1,0,0) + (0,1,0) + (0,0,1) = (0,0,0)$$

over $\mathbb{F}_2$.

Multiply the second row:

$$(0,1,1,0,1)H^T = (1,0,1) + (1,0,0) + (0,0,1) = (0,0,0).$$

Hence

$$\boxed{G'H^T = 0.}$$